

LLM - 2.1 dars.

Data ga qarab preprocessing har
xil bōladi

Datasetni tarkibi (ichki qismi):

- Token IDs
- Sequence
- Batches

Evaluation:

- Loss
- Perplexity
- Generation
- Benchmark

Inference:

- Load checkpoints
- Tokenize prompt
- Generate tokens
- Decode output

Set-up:

- GPU
- VSCode → Google Colab
- Jupyter
- Kernel
- Python 3

TPU → kotta modellar u-n / parametrlar kōp bōlgenda

NVIDIA → learning, debugging, prototyping, small model building

Pretraining: LM model kotta va labellanmagan datani o'rganishi va bu jarayon eng ogir va qimmat hisoblanar ekan

Model tokenlarni o'rganishni / o'qishni boshlaydi, uning asosiy vazifasi next token prediction qilish ekan

Maqsadi: grammatika, faktlarni va bilimni
asosini qurish

Natija: text-predict qiladi, ammo suhbatni
davom ettirishni va savolga javob berishni
bilmaydi.

Post-Training: Modelni toza holatga olib kelish,
raw text predictionlarni chat assistant
ko'rinishiga olib keladi

Post-Trainingda SFT yoki RLHF texnikalaridan
foydalaniladi

Maqsadi: Ko'rsatmalarga amal qilish, toneni
saqlash, alignmentlar qilish

LLM modellar qurayotganda ~~hali~~ eng ko'p vaqt
olara bilan ishlashga ketar ekan.